DeepSkin

INFO9023 - Spring 2025

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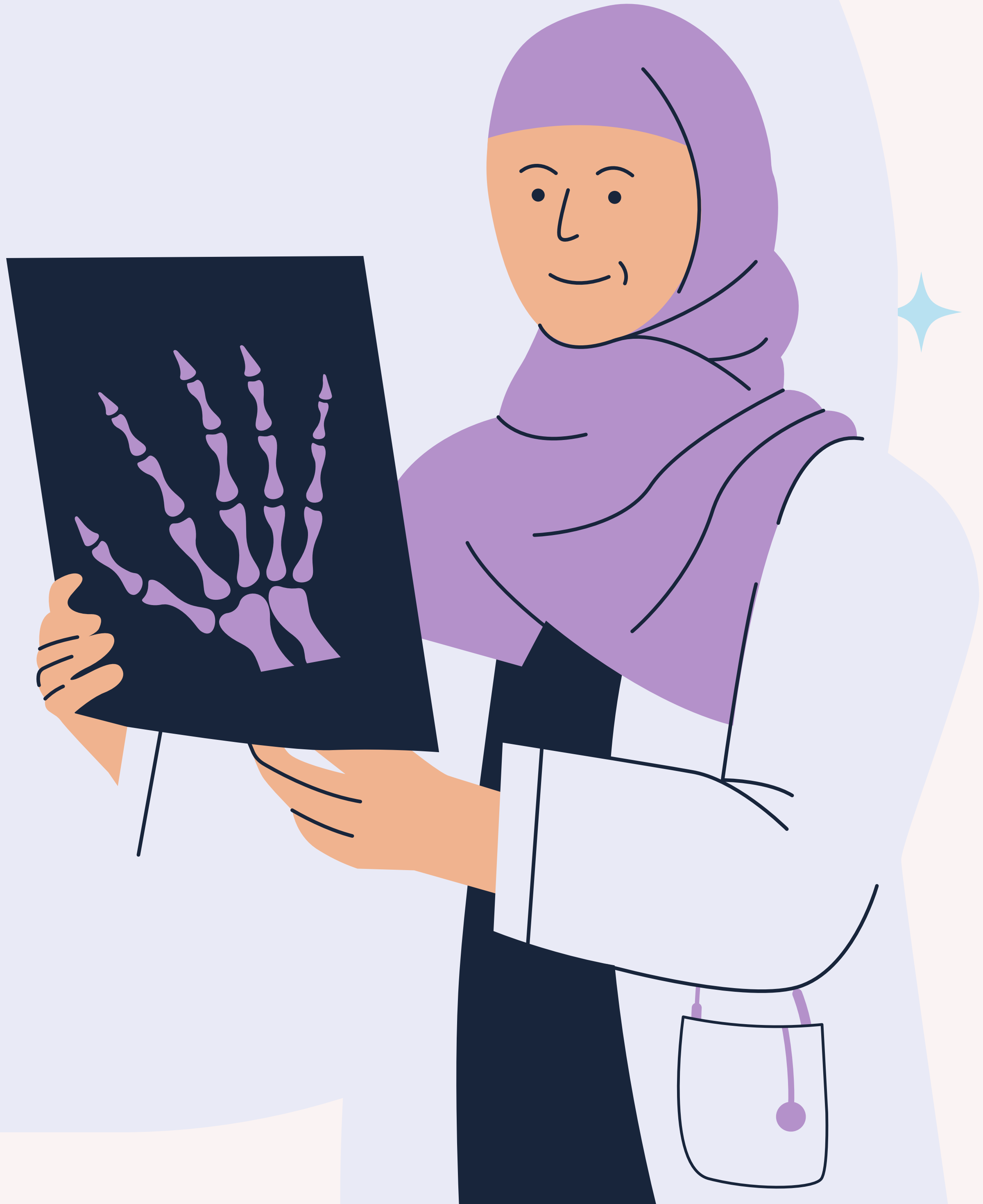
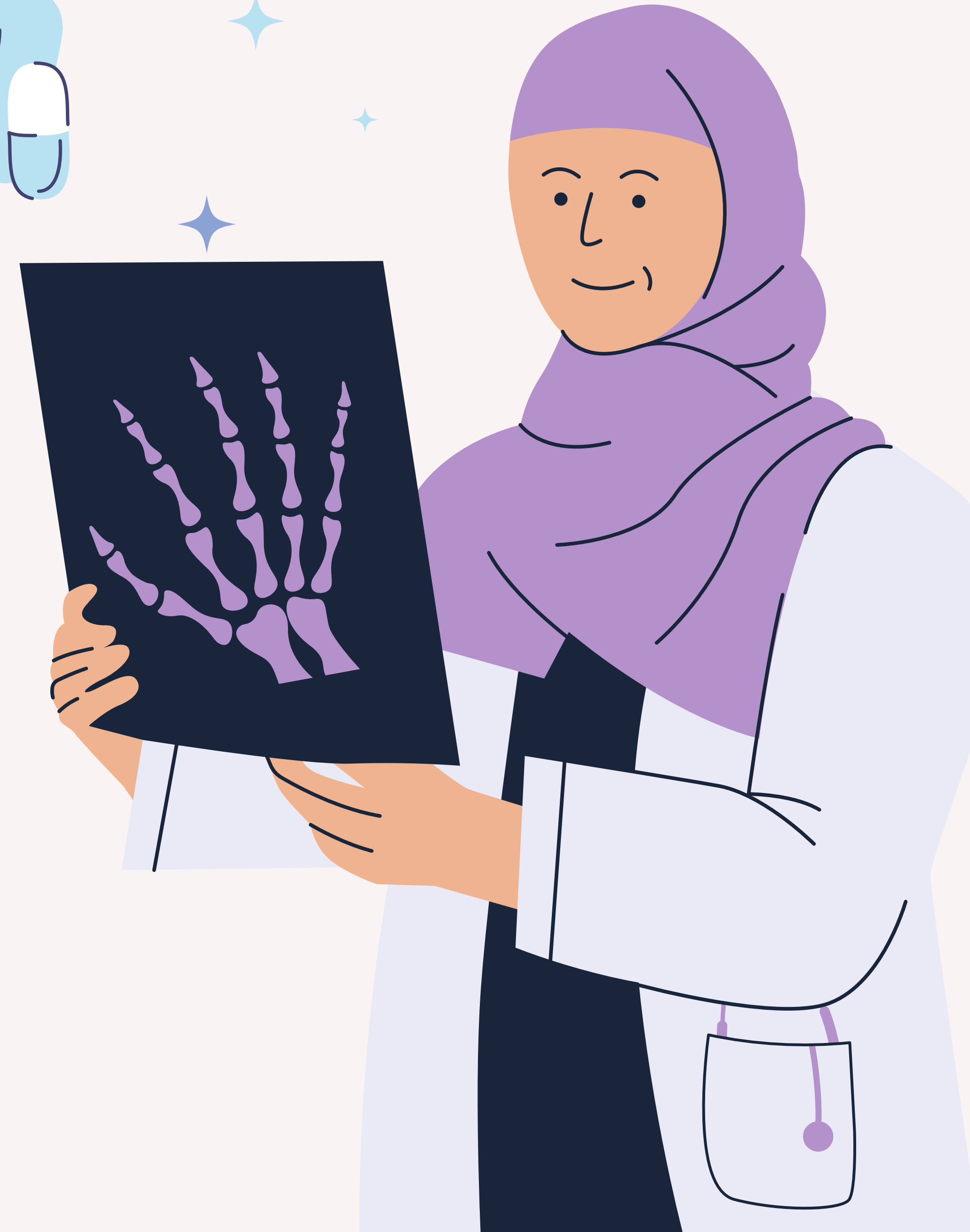
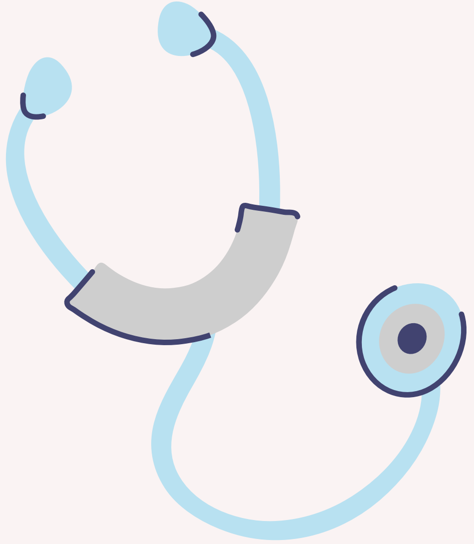


Introduction



Current State of the medical world



Nowadays, obtaining a hospital appointment can take several weeks or even months. However, some cancers and illnesses require **immediate** medical attention, and it is unreasonable to expect medical staff to handle such demand.

To deal with this problem, we propose **DeepSkin** an ML-based solution designed to reduce the burden on healthcare providers by filtering **non-urgent** skin conditions from those that require immediate attention.

Machine Learning Canvas

Value Proposition

“We propose an application for automatically identifying nature and potential risk assessment of user's lesions. “

Objectifs

1. Provide the user with a interactive and easy to use interface
2. Classify the nature of the lesion, and give a probability score on the prediction

Solution

The idea behind our solution is for the users to upload their skin problems on our application and they receive a predicition as well as a probability score associated to it

Feasability

1. Regarding the data, we found the dataset
2. we all are familiar with deep learning methods
3. For the infrastructures we will use modern tools such as Google Cloud, Vertex, ...

Dataset

DataSet

The HAM10000 DataSet

kaggle

This dataset is a large collection of multi-source dermatoscopic images of pigmented lesions composed of 3 files

- The first one is a **metadata** file with the localization of the image, the type of the lesion as well as personal data such as the **age** / **sex** / **localization** of the lesion

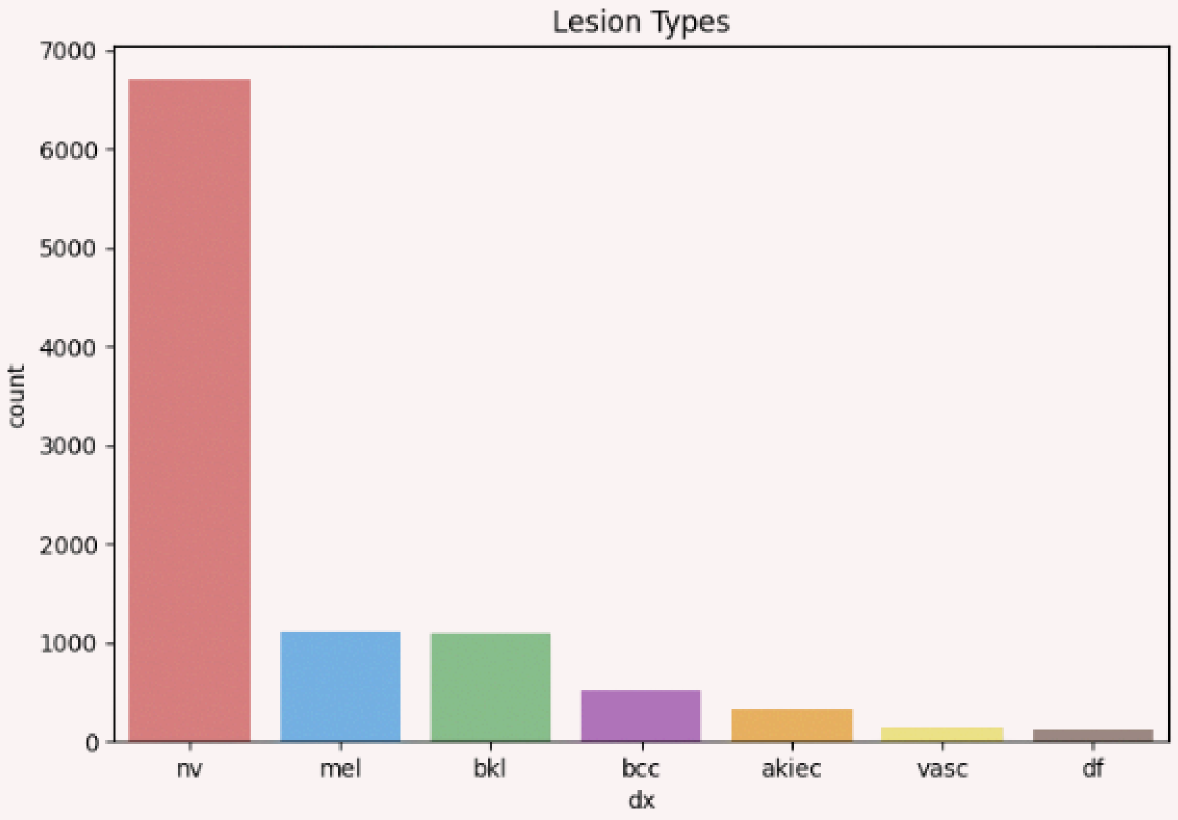
- The 2 other file are the image folders each of them containing 5000 skin lesions pictures

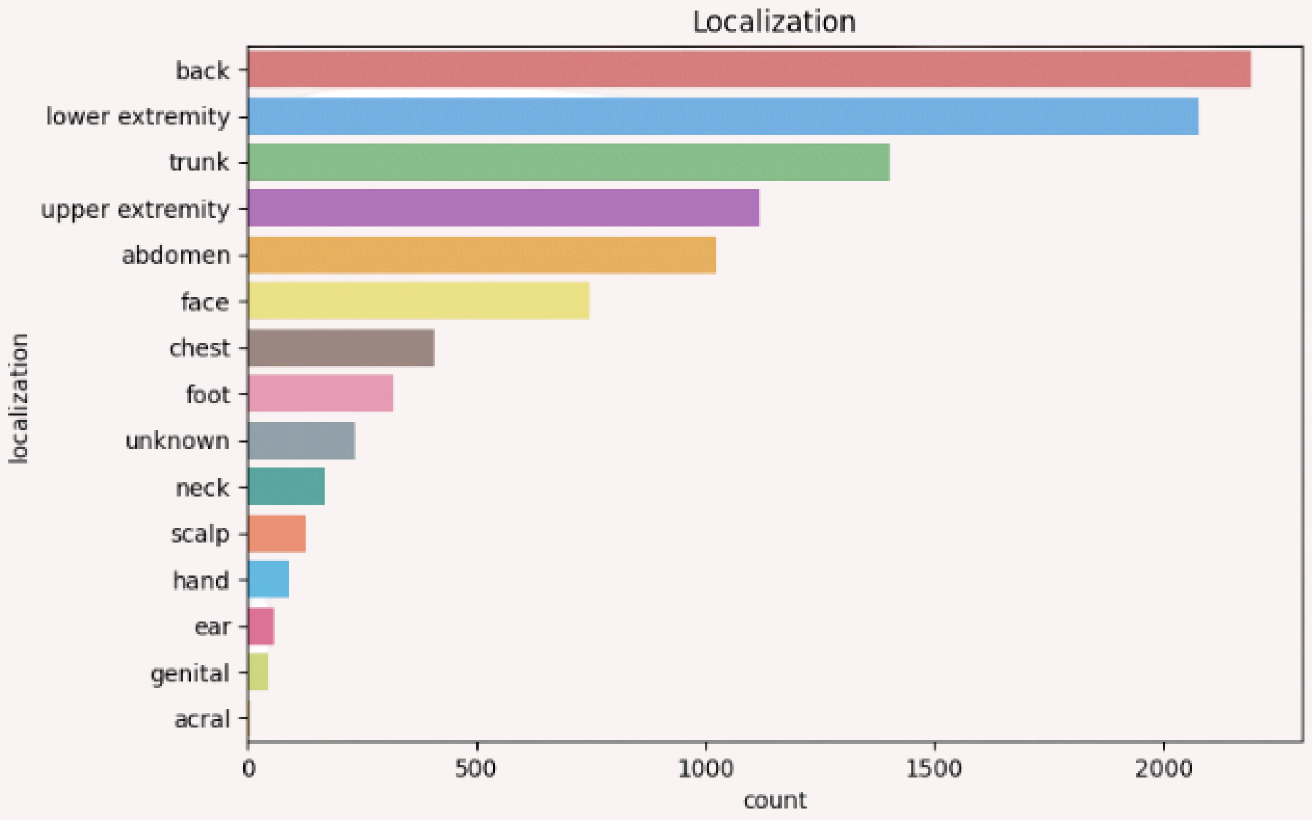
DataSet

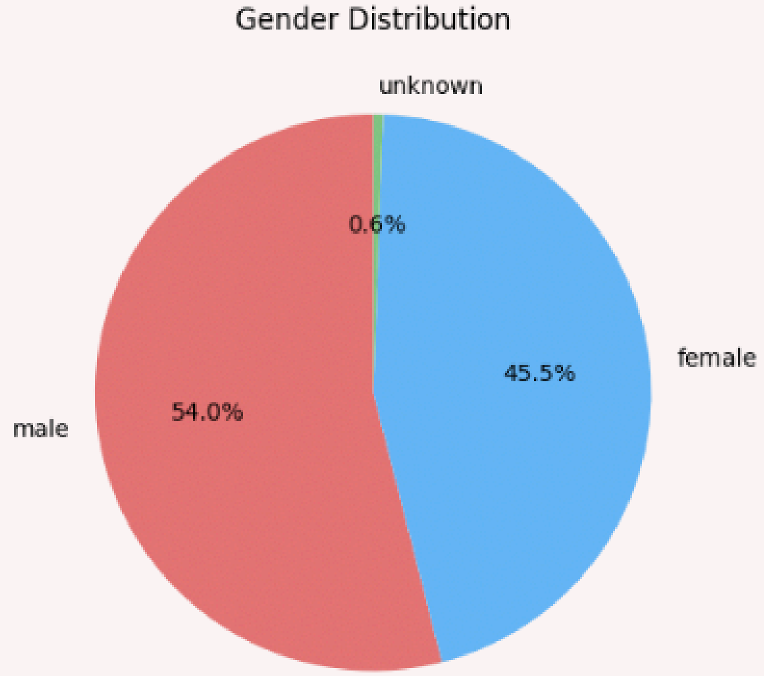
Exploratory Analysis

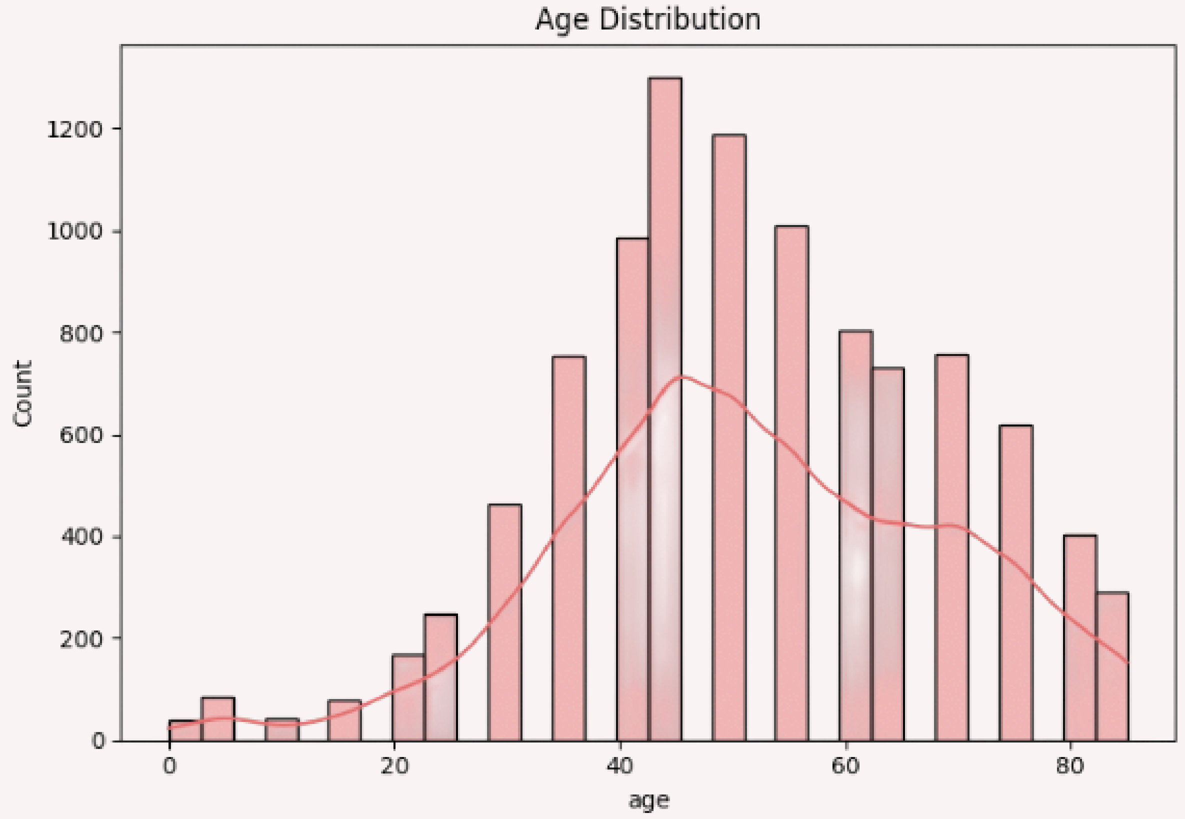
Data Distribution

We explore how the data was distributed among different metrics in the dataset as we can see there is a huge imbalance regarding the nv class







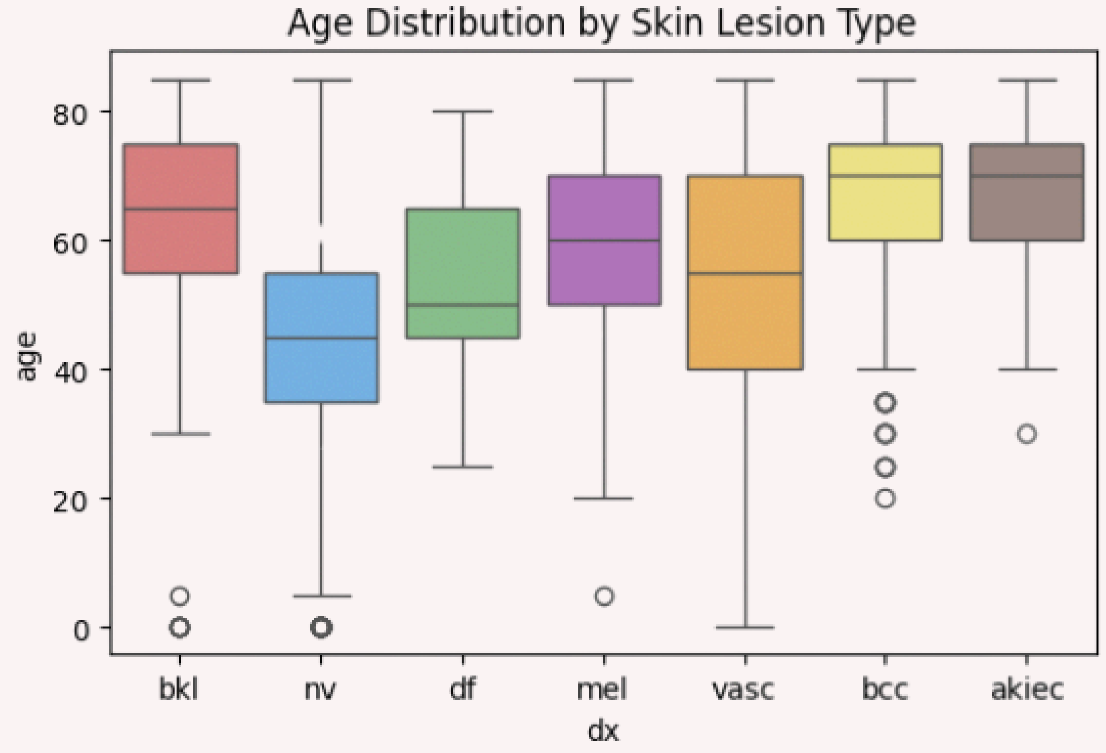


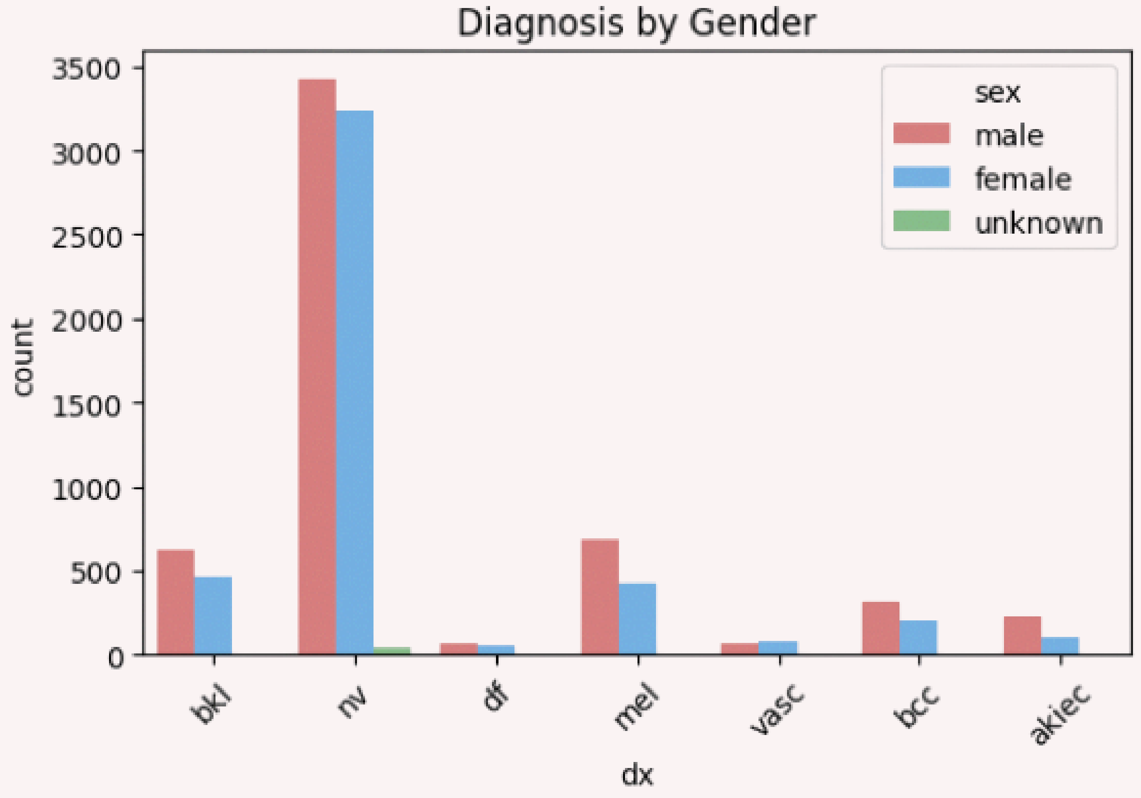
DataSet

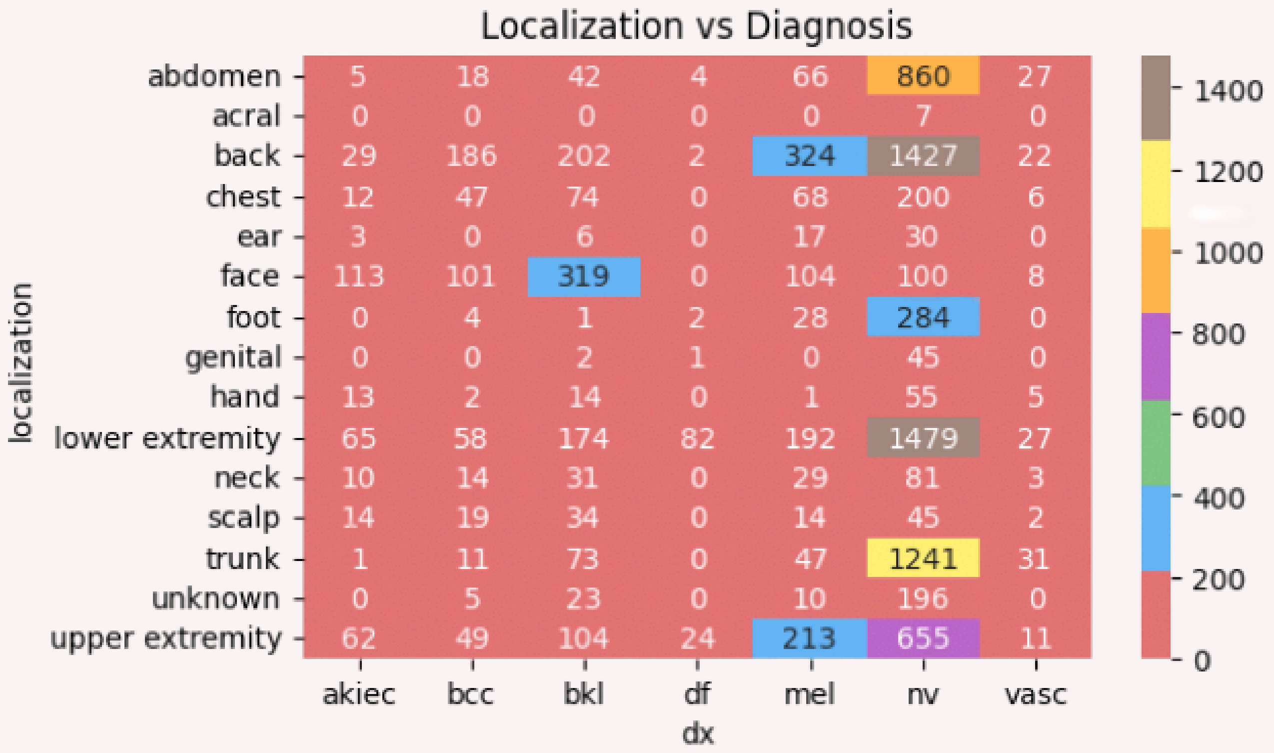
Exploratory Analysis

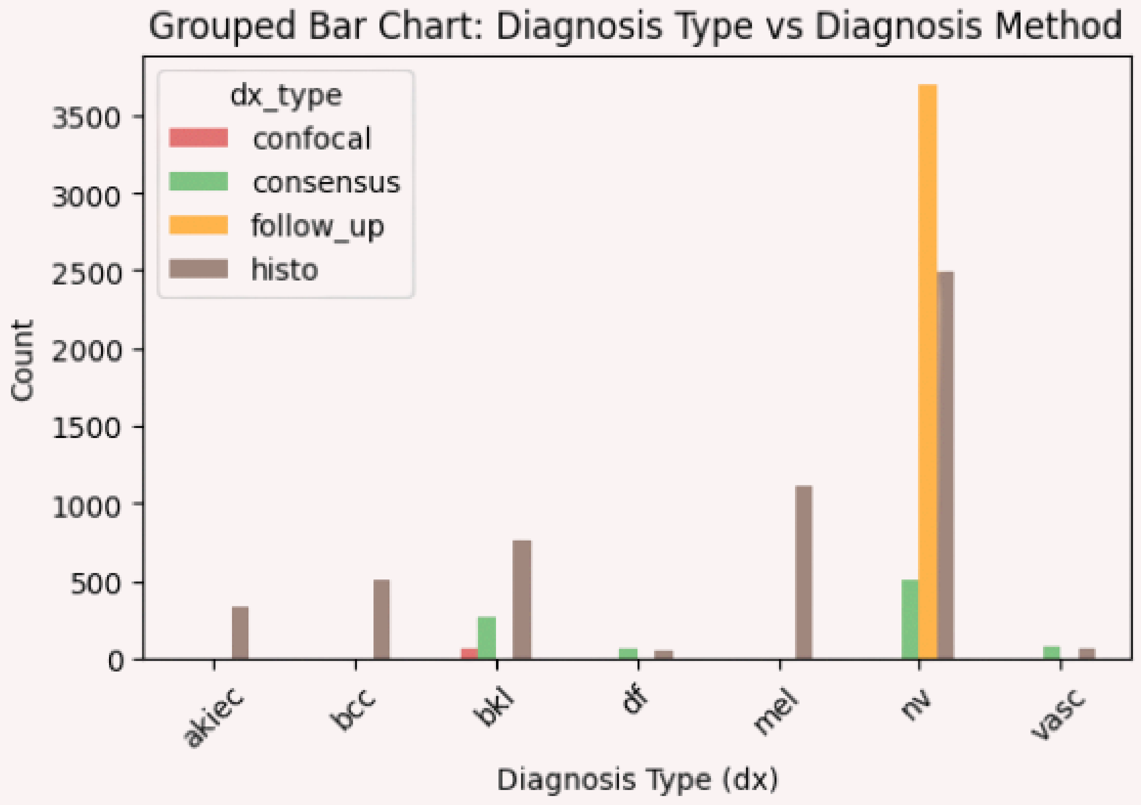
More Analysis

We also explore other relation among our dataset such as age distribution for some type of lesion, the gender distribution etc..









Model

Model Architecture

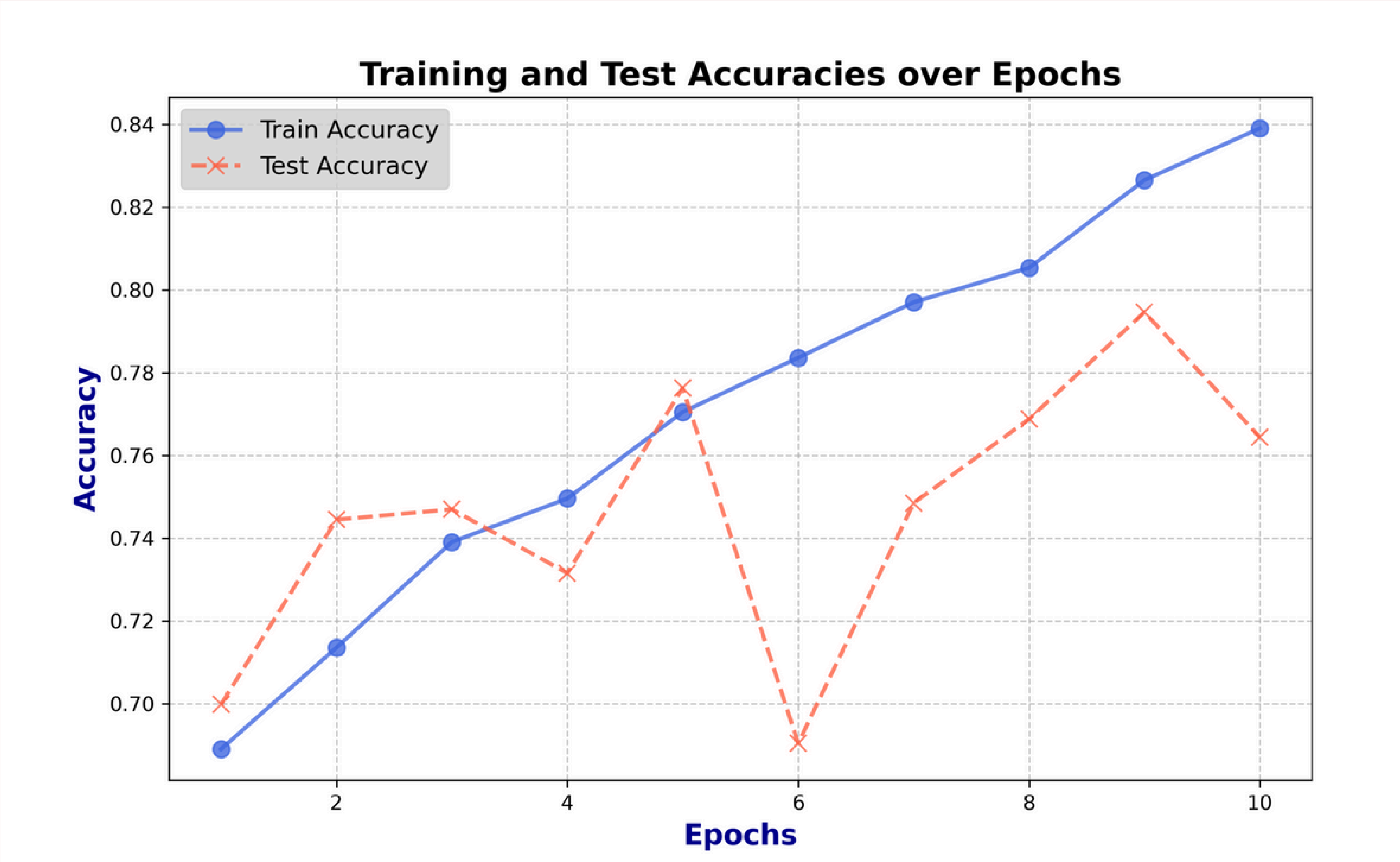
To use as well as we can all the information available in the dataset we will not only use the image and lesion type but also the age / sex and localization as metadata for the prediction

Regarding the image we use a pre-trained model , **ResNet 18**, to extract the features of the image. On the side the metadata are pass through a basic **MLP** with 2 hidden layers

Finally we concatenate both results and inject that into a **classifier** to make our predictions

Model

Model Monitoring



Next Step FAQ

Our Next Step / Improvements

Data Augmentation

As said we will use data augmentation to change our dataset, to avoid such a huge class imbalance

Deployment

We will start the deployment of our model on the cloud

Bash Script

We will simplify our life by creating batch script to pass Hyperparameters value directly on the run command

WandB Monitoring

We will also integrate WandB into our codebase to monitore the performances of our model as well as adding more metrics to access its quality

Thank you for your attention

Any Questions ?